

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
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Experiment 3: Regression Analysis using Linear and Regularized Models

1. Aim and Objective

The primary objective of this study is to construct and evaluate multiple regression architectures, specifically linear, Ridge, Lasso, and Elastic Net models, to forecast a continuous target variable. The experiment seeks to analyze predictive accuracy across various metrics and investigate the trade-offs between bias and variance through hyperparameter optimization.

2. Dataset

The investigation utilizes a real-world financial dataset containing diverse numerical and categorical attributes related to loan applications. The goal is to accurately predict the **loan amount sanctioned** for each applicant based on their profile.

3. Preprocessing Steps

A comprehensive data transformation pipeline was established to ensure that the input features were properly formatted for linear modeling.

Handling Missing Values

Numerical Data

Missing entries in numerical columns were addressed using a *median* imputation strategy, which provides a measure of central tendency that is less sensitive to extreme outliers.

Categorical Data

For categorical features, missing observations were replaced using the *most-frequent* value (mode) of the respective column to maintain the categorical distribution.

Encoding Categorical Variables

Categorical variables were transformed into a machine-readable format via **One-Hot Encoding**. The encoder was configured to ignore unknown categories during the testing phase to prevent execution errors.

Feature Scaling

Standardization was performed on all numerical features using **StandardScaler**, centering the data at a zero mean and scaling to unit variance. This ensures that regularization penalties are applied equitably across features with different original scales.

4. Implementation Details

The study compared a baseline linear model against several regularized variants to determine the most effective approach for this dataset.

Regression Models

Linear Regression (Baseline)

The baseline model was established using standard Ordinary Least Squares (OLS) without any regularization penalties.

Ridge Regression (L2 Regularization)

Ridge models were implemented to prevent coefficient explosion by adding a squared magnitude penalty ($L2$). The tuning grid explored $\alpha \in \{0.01, 0.1, 1, 10, 100\}$.

Lasso Regression (L1 Regularization)

Lasso regression was utilized to encourage feature sparsity through an absolute magnitude penalty ($L1$). The grid search evaluated $\alpha \in \{0.001, 0.01, 0.1, 1, 10\}$.

Elastic Net Regression

This hybrid model combined both $L1$ and $L2$ penalties. The hyperparameters tuned included $\alpha \in \{0.01, 0.1, 1, 10\}$ and $l1_ratio \in \{0.2, 0.5, 0.8\}$.

Validation Strategy

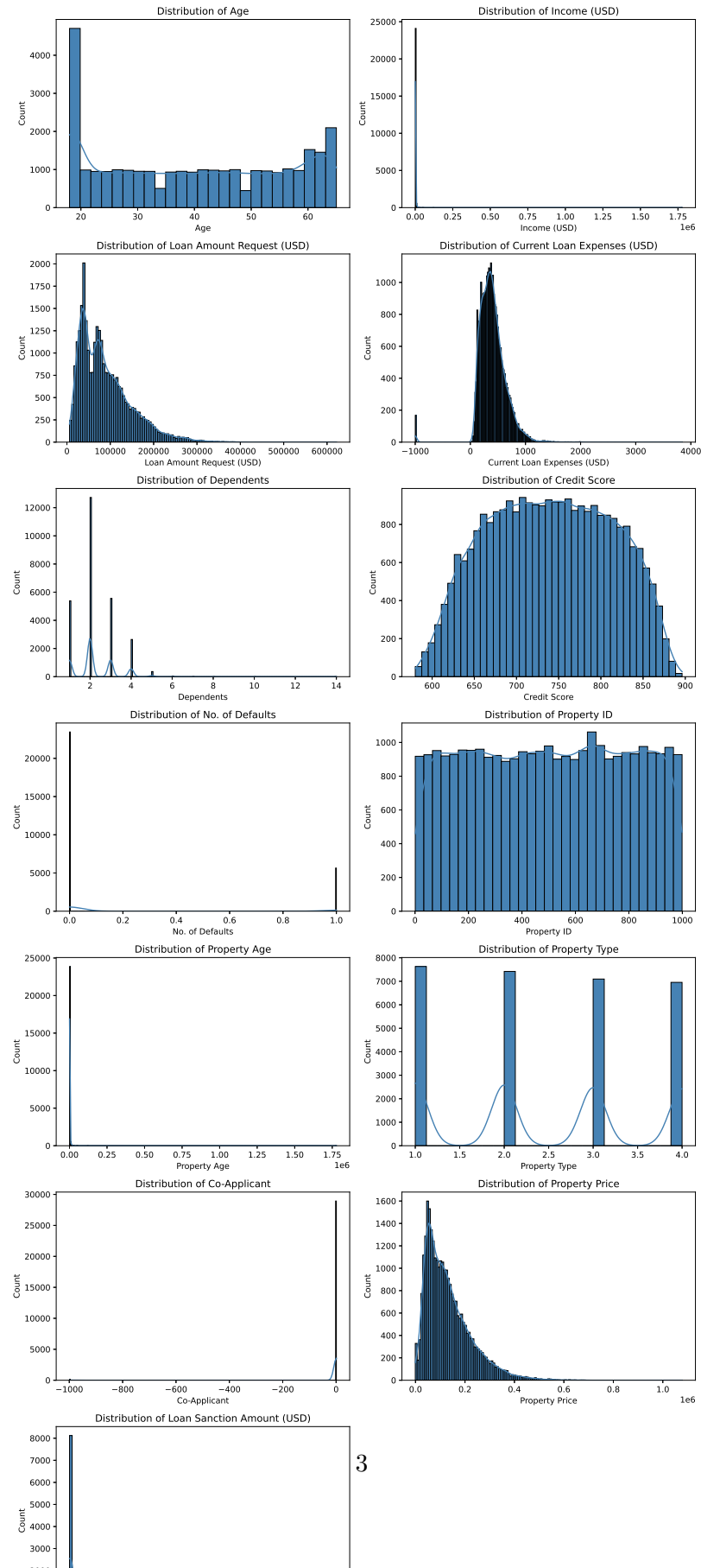
K-Fold Cross-Validation

A 5-Fold Cross-Validation approach was used to ensure that the selected hyperparameters provided stable performance across different data subsets.

Scoring Metric

The models were optimized based on the R^2 score (Coefficient of Determination), representing the proportion of variance captured by the model.

5. Visualizations



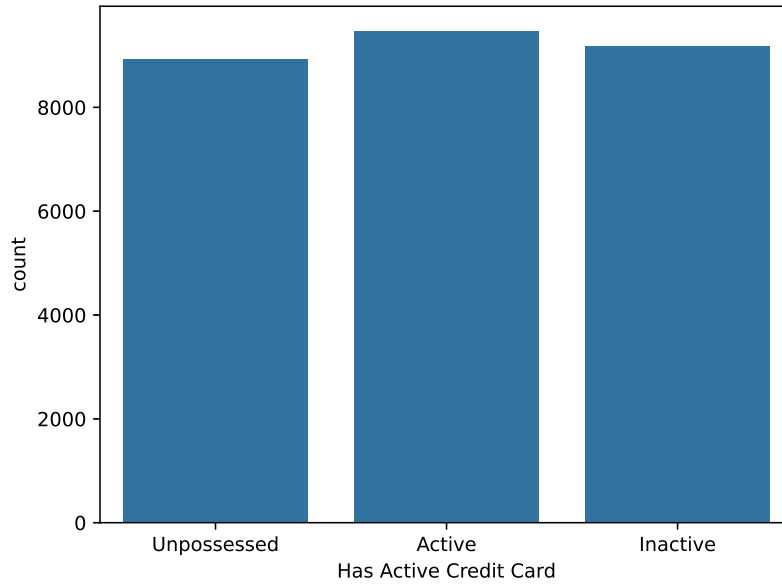


Figure 2: Depicting Missing Values

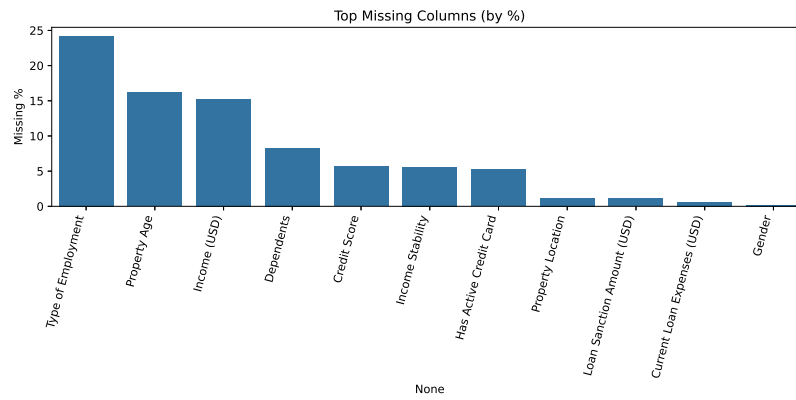


Figure 3: After filling missing Values

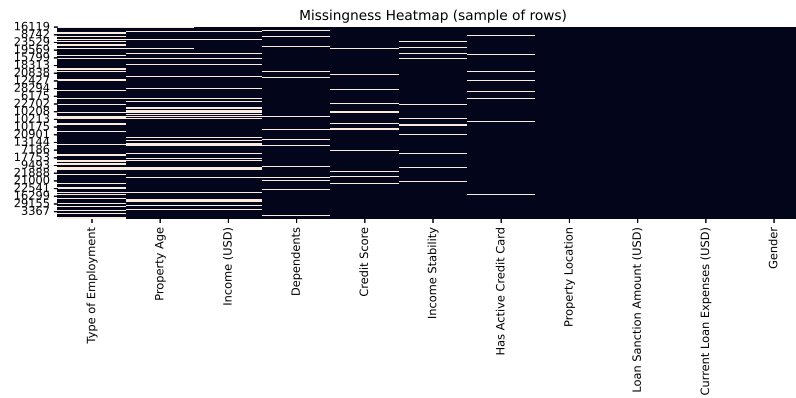
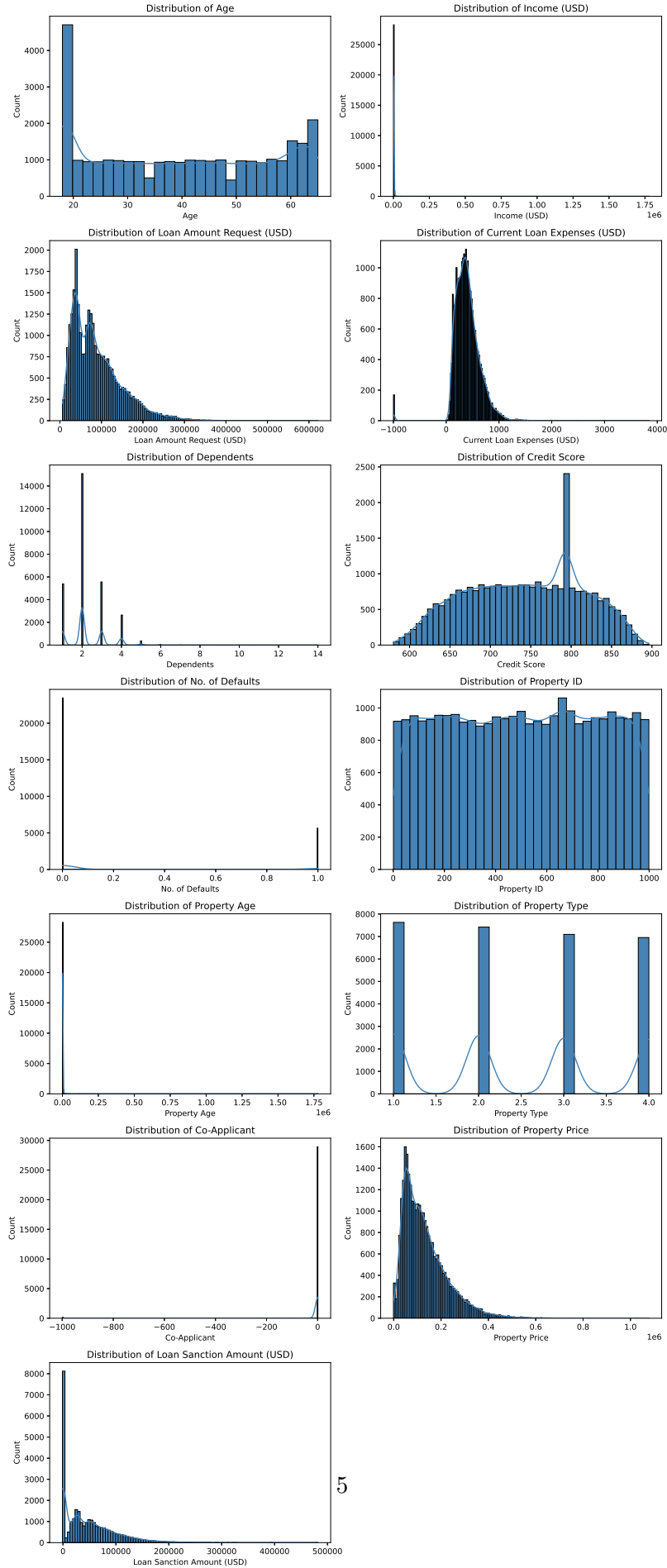
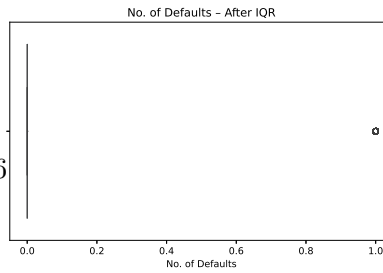
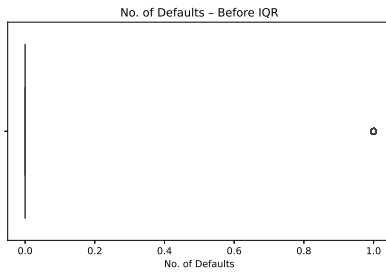
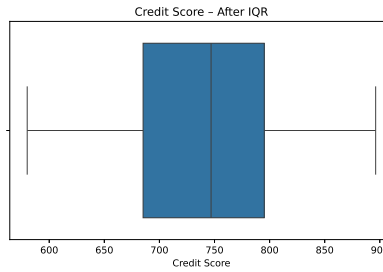
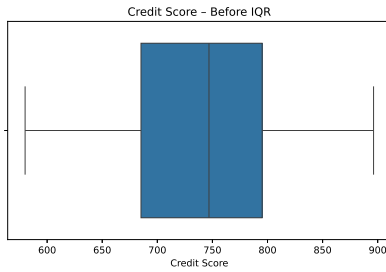
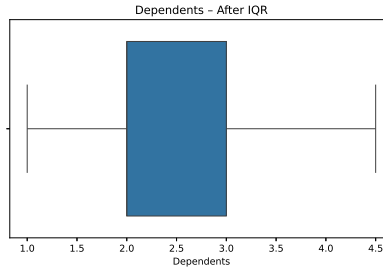
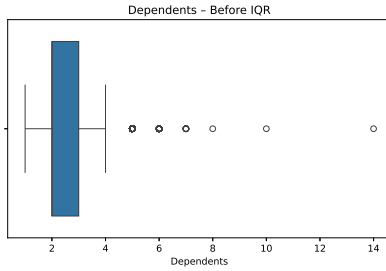
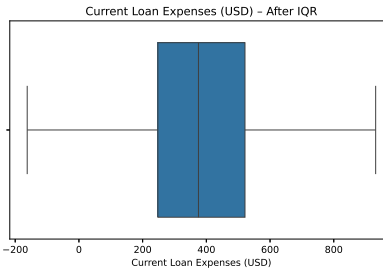
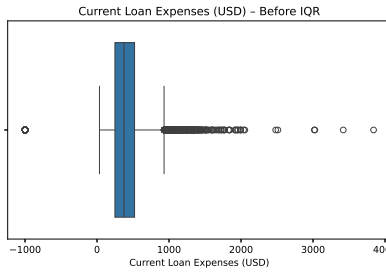
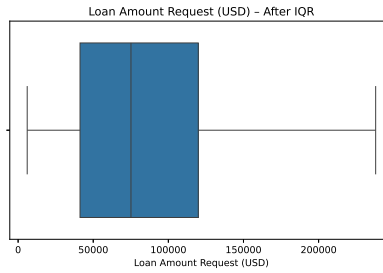
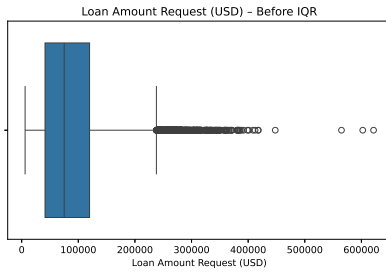
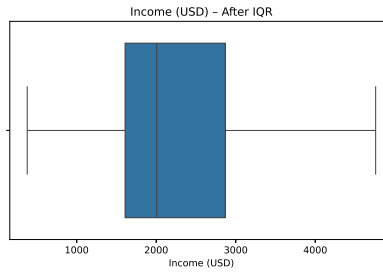
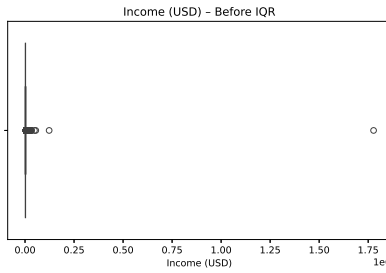
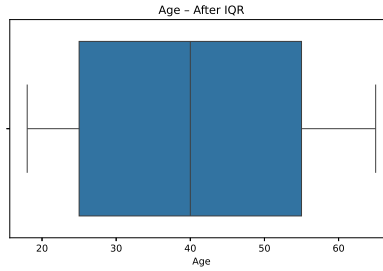
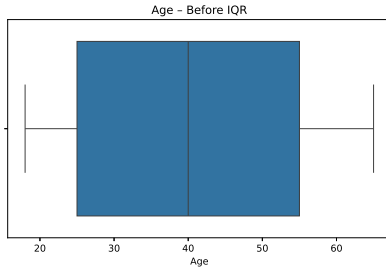
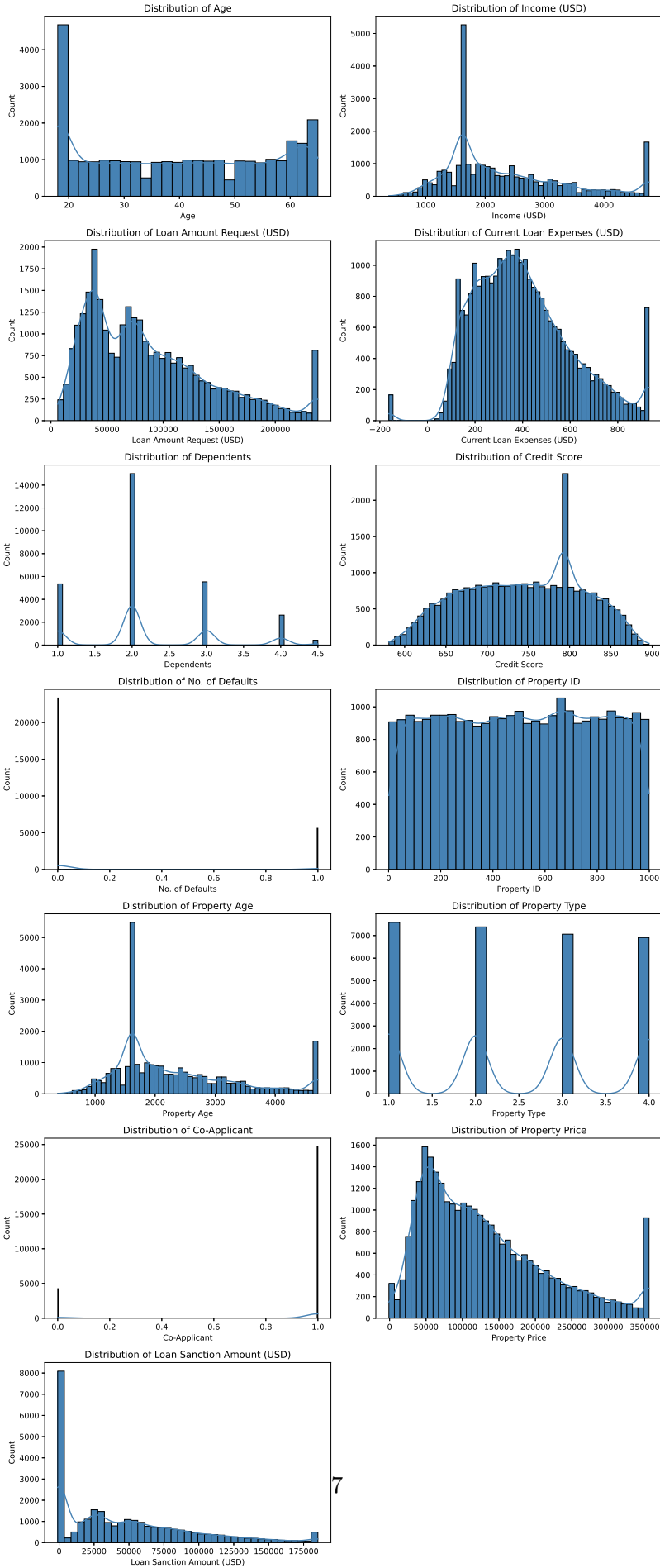
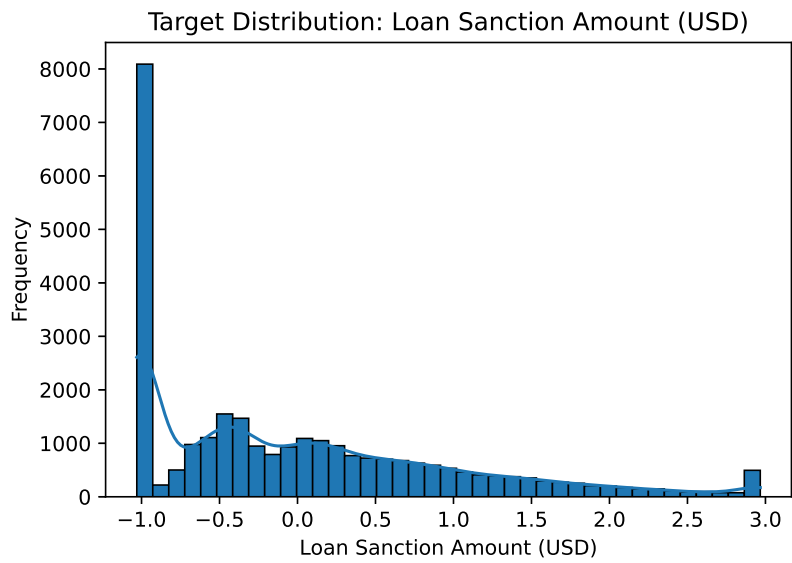
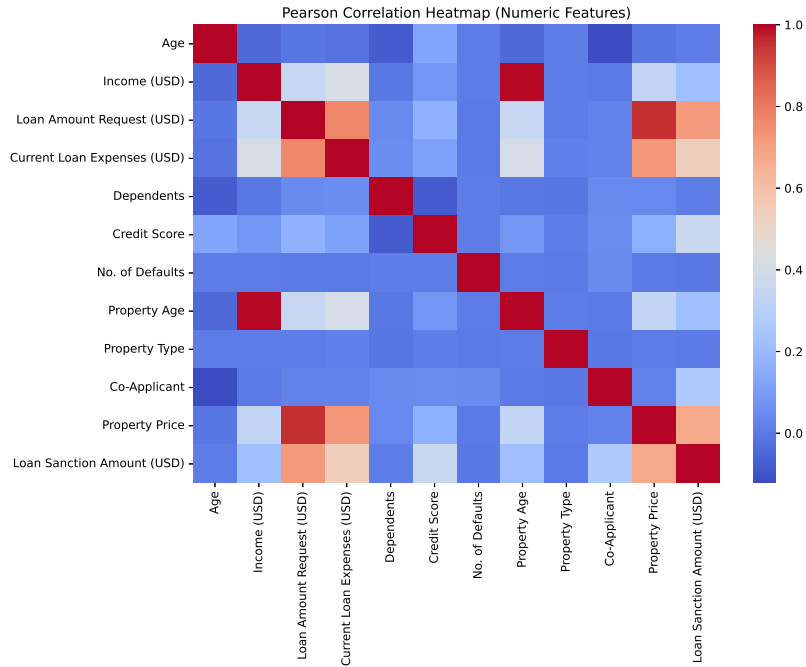


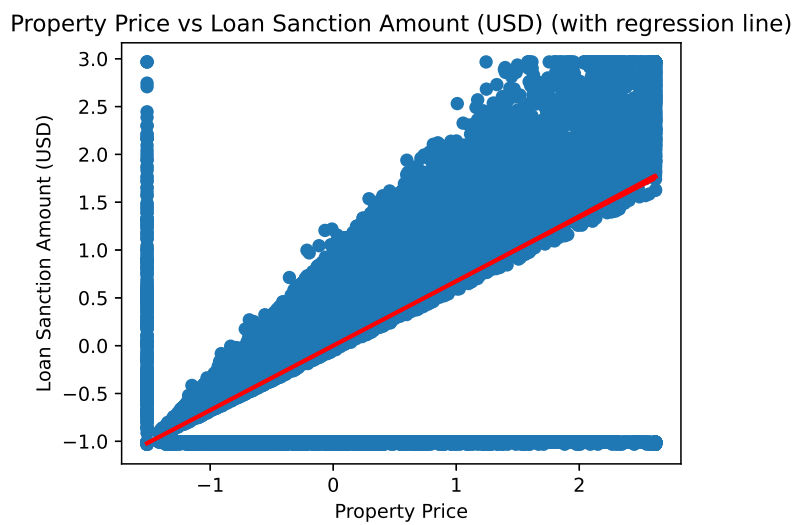
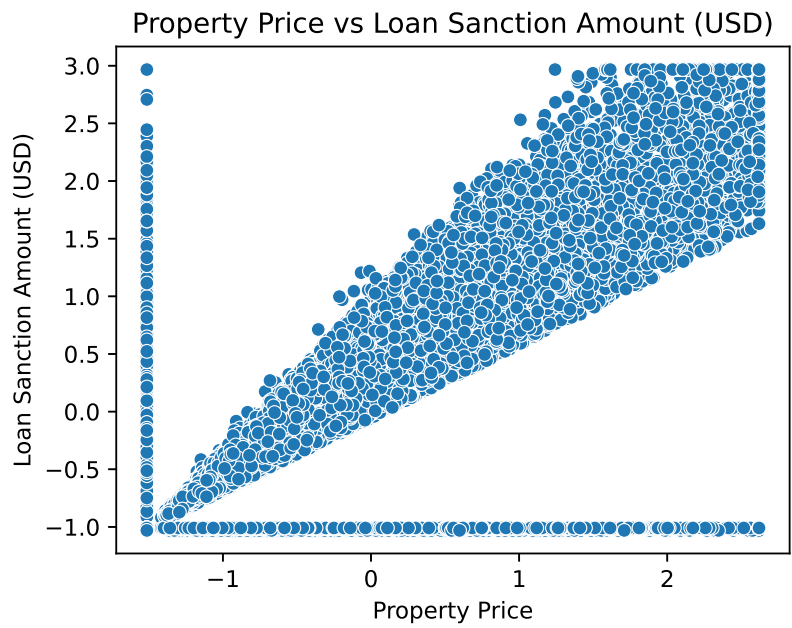
Figure 4: Distribution after Outlier reduction

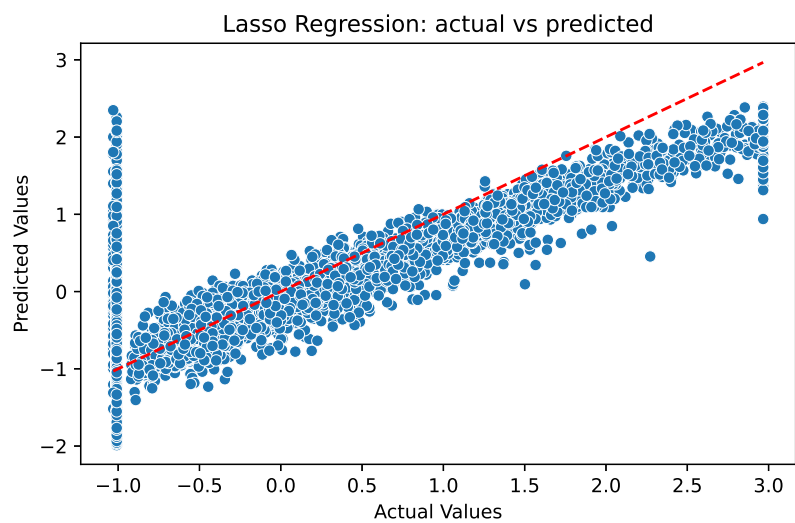
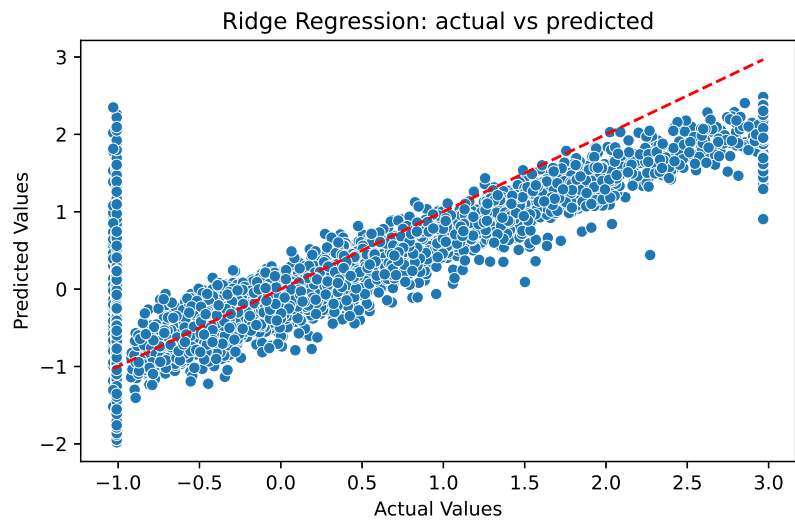
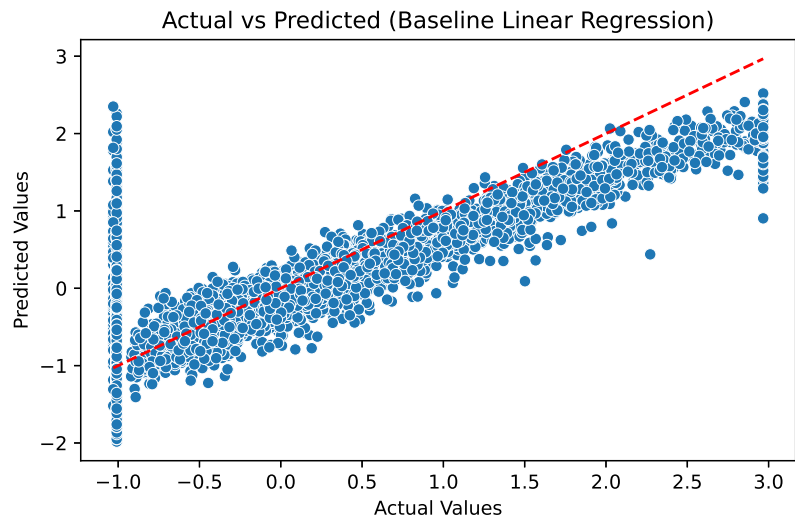


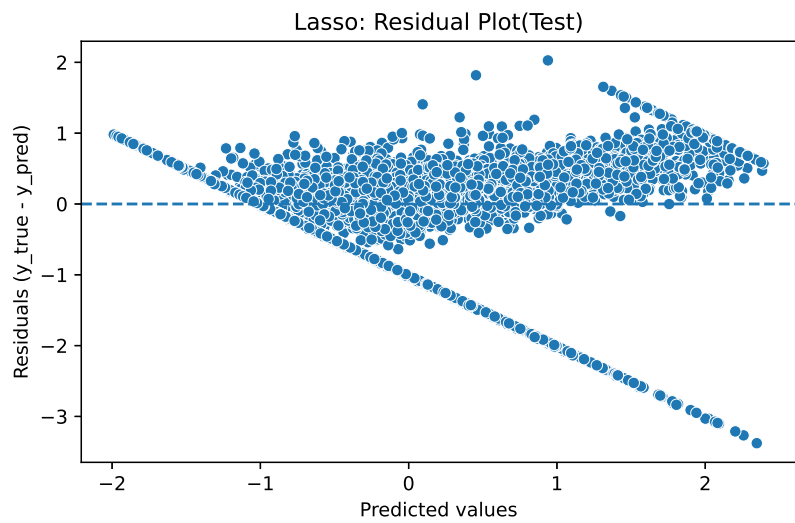
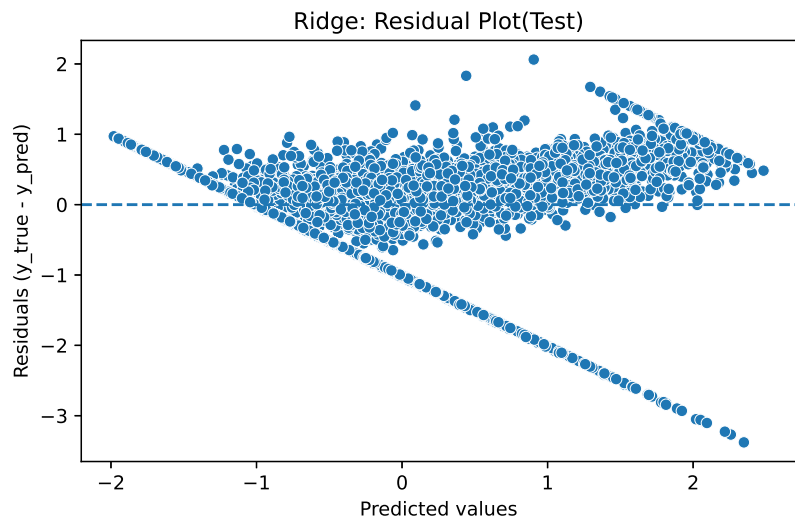
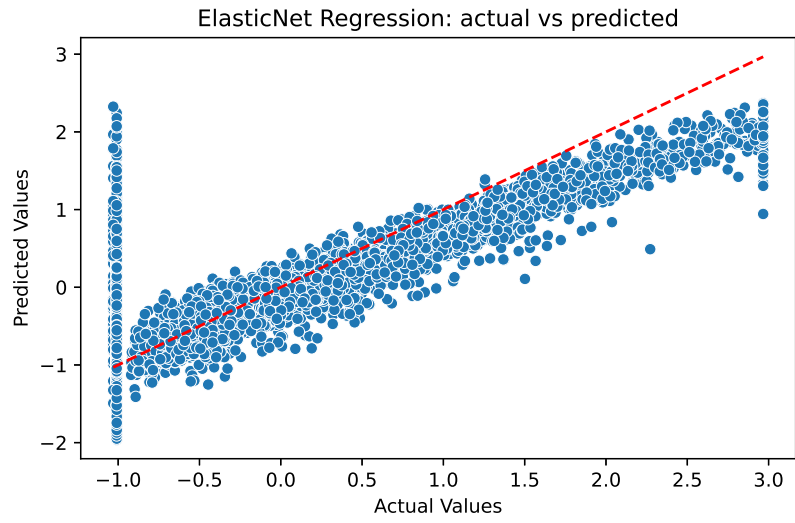


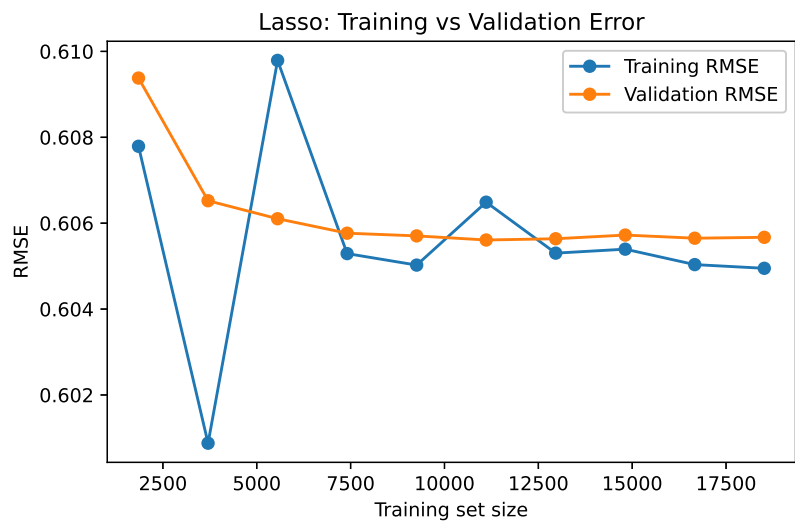
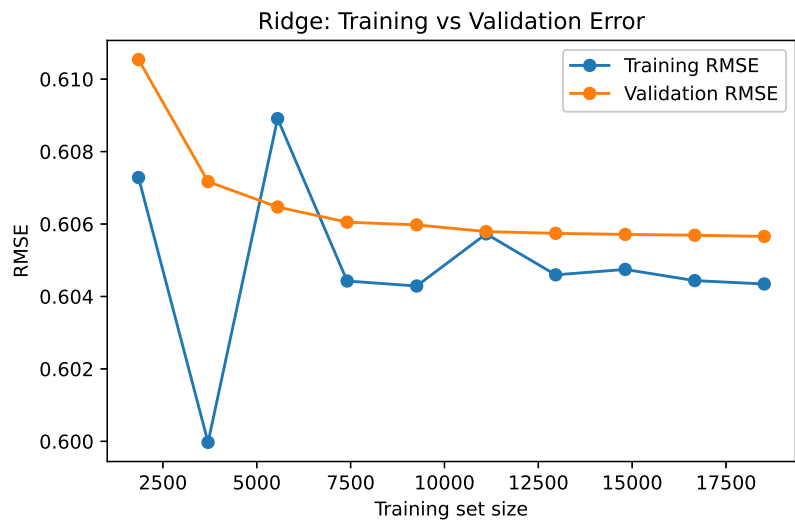
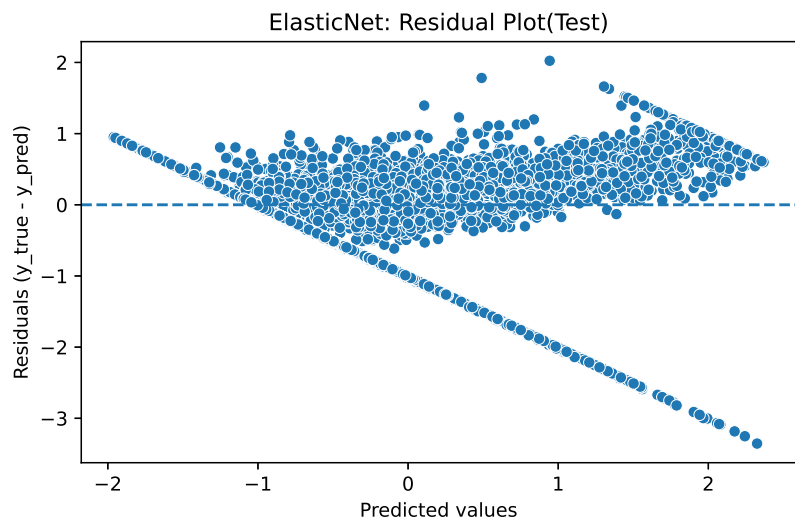


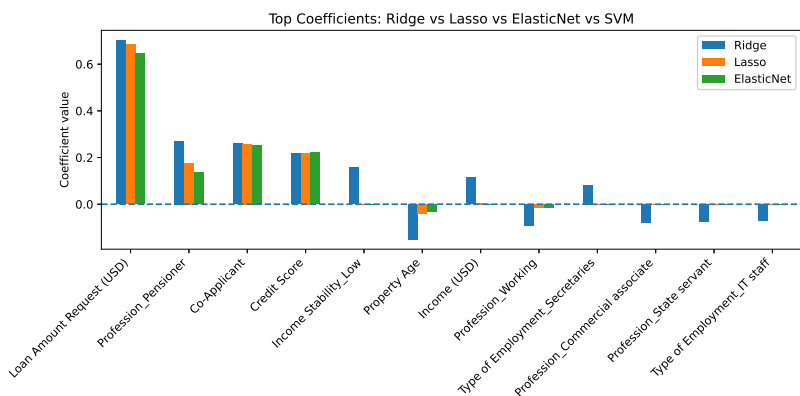
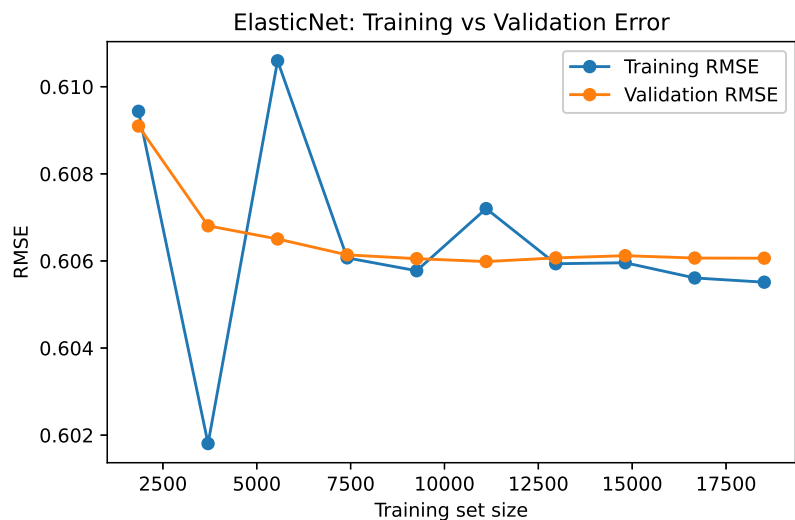












6. Performance Tables

Hyperparameter Tuning Results

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid	alpha: 10	0.6317
Lasso Regression	Random	alpha: 0.001	0.6316
Elastic Net Regression	Random	alpha: 0.01, l1_ratio: 0.2	0.6311

Table 2: Cross-Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.4161	0.3668	0.6057	0.6316
Ridge Regression	0.4160	0.3668	0.6056	0.6318
Lasso Regression	0.4155	0.3668	0.6056	0.6316
Elastic Net Regression	0.4162	0.3673	0.6060	0.6311

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.4149	0.3623	0.6019	0.6436
Ridge Regression	0.4149	0.3623	0.6019	0.6435
Lasso Regression	0.4152	0.3628	0.6024	0.6430
Elastic Net Regression	0.4162	0.3636	0.6030	0.6423

Table 4: Coefficient Comparison

Feature	Linear	Ridge	Lasso	Elastic Net
Loan Amount Request	0.7060	0.7024	0.6833	0.6465
Co-Applicant	0.260458	0.260225	0.258592	0.253734
Credit_score	0.21727	0.217431	0.220091	0.221885

7. Overfitting and Underfitting Analysis

Training vs. Validation Gap

The cross-validation R^2 scores, remain Relatively stable, around the 0.6316 mark. Hence there is little overfitting taking place.

Underfitting Analysis

Despite the stability of the models, the test R^2 score of roughly 0.6318 (Ridge) indicates that the models account for only about 63% of the variance in the target variable. This level of performance suggests a degree of underfitting. The linear assumptions of these models appear insufficient to capture the more complex, non-linear interactions likely present in loan sanctioning data, such as the relationship between credit history and final approved amounts.

Effect of Regularization

The variance in test R^2 scores between the baseline Linear Regression (0.6316) and the best-performing regularized model, Ridge Regression (0.6318), is statistically negligible. This confirms that high variance was not the primary hindrance to performance; rather, the models were limited by their representational capacity (bias).

8. Bias–Variance Analysis

Bias Behavior

The high bias inherent in linear models is evident here, as they fail to capture the full complexity of the financial dataset. This results in a relatively high Mean Absolute Error (MAE) of approximately 0.4160 for the baseline model, indicating consistent, systematic prediction errors.

Variance Reduction

- The Linear Regression model assigns relatively large positive or negative coefficients to several features, indicating no constraint on coefficient magnitude. Ridge Regression shrinks these coefficients toward zero while preserving all features, showing the stabilizing effect of L2 regularisation. Lasso Regression drives many profession-related and income stability features exactly to zero, demonstrating its built-in feature selection capability. Elastic Net shows intermediate behavior by shrinking coefficients while retaining a small contribution from correlated features such as Credit Score, combining the strengths of both Ridge and Lasso. .

9. Observations and Conclusion

Feature Importance

The analysis indicates that *Loan Amount Request (USD)*, Co-applicant and *Credit Score* serve as the primary predictors in this dataset. Correlation metrics further support this, showing that the requested amount maintains the strongest positive relationship with the final sanctioned amount.

Model Performance

All four regression architectures performed similarly, with Ridge Regression providing the marginal lead on the test set with an R^2 of 0.6318. The consistent performance across all models suggests that the predictive power of linear relationships for this specific dataset has reached its ceiling.

Conclusion

While regularization proved highly effective at stabilizing model parameters and preventing coefficient "explosion," it could not overcome the fundamental underfitting caused by the linear model constraint. To achieve higher predictive accuracy, future iterations should consider non-linear approaches such as Random Forests, Support Vector Regression (SVR), or Gradient Boosting.

Dataset reference

- Kaggle: Predict Loan Amount Data

References

- Scikit-learn: Linear Models
- Scikit-learn: Hyperparameter Optimization

- Loan Amount Dataset